

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech. (CE) Examination****March-April 2018****CE243 Data Communication and Networking****Date: 02.04.2018, Monday****Time: 01.30 pm to 04.30 pm****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below. [11]**

- [A] What is composite signal? Give one example. [2]
- [B] Which layer of OSI model is responsible for presentation of data? [3]
- [C] What is length of port address, a logical address, and a physical address? [3]
- [D] For effective data communication, what must the fundamental characteristics? [3]

Q - 2 Answer any THREE. [12]

- [A] Write down the responsibilities of Network layer of OSI model.
- [B] Explain 10Base5, 10Base2, 10BaseT and 10BaseF.
- [C] List several transmission media for networking. Explain any two media in brief.
- [D] What is Multiplexing? Explain Frequency division multiplexing.
- [E] In terms of data transmission, what is Broadcast, Multicast, Unicast and Peer-to-Peer communication?

Q - 3 Answer any TWO. [12]

- [A] Write down different topologies of communication network. Also, write their advantages and disadvantages.
- [B] Draw and explain functionality of each layer of OSI model.
- [C] Explain guided media and unguided media with example.
- [D] Differentiate LAN, MAN and WAN.

SECTION – II**Q - 4 Answer the question below. [11]**

- [A] How does a single-bit error differ from a burst error? [2]
- [B] Write down classification of various flow control protocols. [3]
- [C] Define following terms [3]
(i) Wavelength (ii) Amplitude (iii) Propagation Delay
- [D] What do you mean by Medium Access Control? [3]

Q - 5 Answer any THREE. [12]

- [A] Write down the differences between circuit switching and packet switching network.
- [B] Explain VRC with example.
- [C] Write a short note on Modem.
- [D] Write the differences between serial and parallel communication.
- [E] What do you mean by DTE/DCE interface? Explain in short.

Q - 6 Answer any TWO. [12]

- [A] Draw and explain Ethernet 802.3 frame format.
- [B] Write down the functionalities of following devices
Repeaters, Hub, Bridges , Switches, Routers, Gateways
- [C] Explain Analog-to-Analog conversion by taking suitable example.
- [D] What is routing algorithm? List different types of routing algorithm. Explain link state algorithm with example.
